

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 4: Equilibrium of a Particle

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Condition For the Equilibrium of a Particle

A particle is in equilibrium if it remains at rest if originally at rest or has a constant velocity if originally in motion. To maintain equilibrium, it is necessary to satisfy **Newton's first law** of motion, which requires the resultant force acting on a particle to be equal to zero.

$$\Sigma \mathbf{F} = \mathbf{0}$$

The Free-Body Diagram

To apply the equation of equilibrium, we must account for all the known and unknown forces ($\Sigma \mathbf{F}$) which act on the particle.

The best way to do this is to make a drawing of the particle showing the particle **isolated** or “**free**” from its surroundings. When this drawing shows all the forces that act on the particle, it is called a free-body diagram (**FBD**).

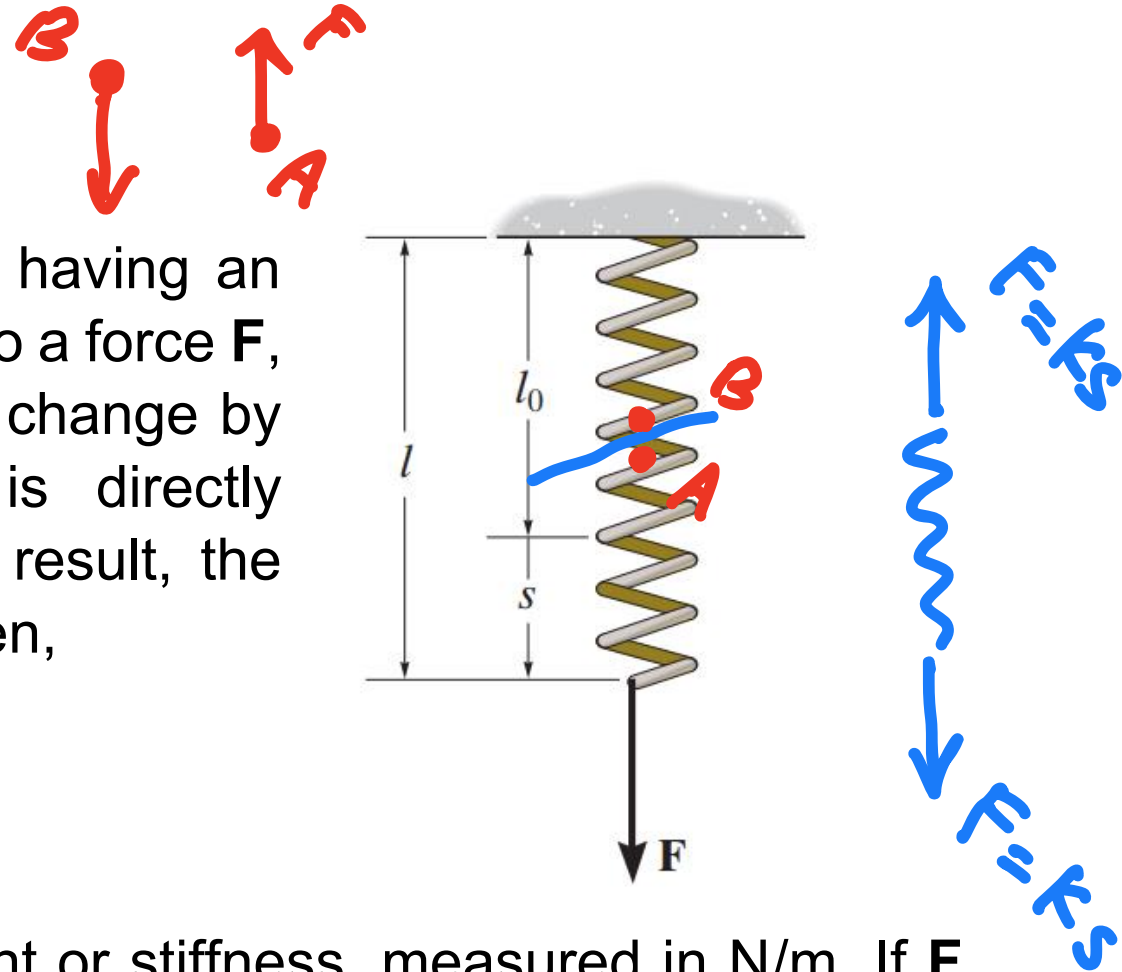
The Free-Body Diagram

Types of supports: Springs.

If a linear elastic spring (or cord) having an undeformed length l_0 is subjected to a force $\mathbf{F}$, then the length l of the spring will change by an amount $s = (l - l_0)$ that is directly proportional to the force $\mathbf{F}$. As a result, the relationship between F and s is then,

$$F = ks$$

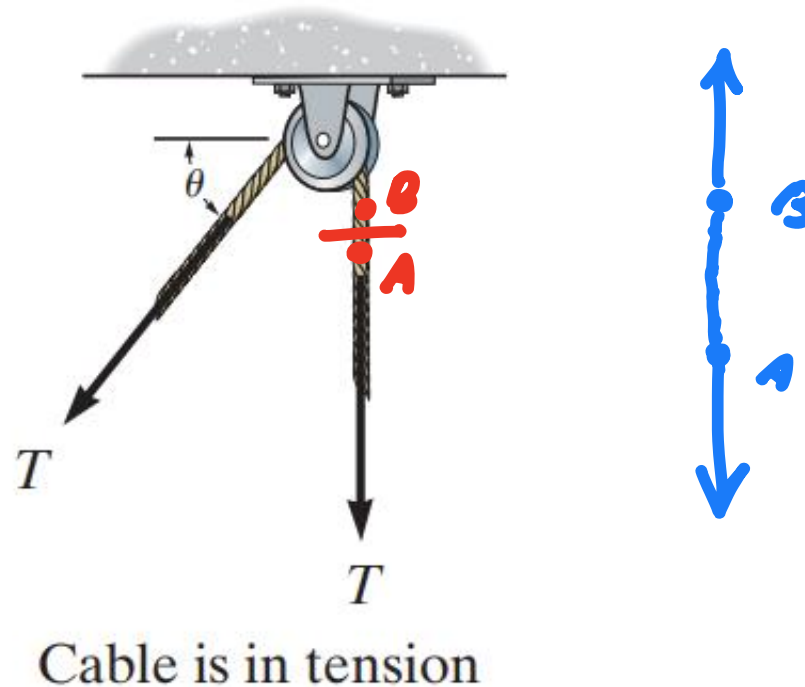
Here k is called the spring constant or stiffness, measured in N/m. If $\mathbf{F}$ pulls on the spring, then s represents an **elongation**, whereas if $\mathbf{F}$ pushes on the spring then s is a **contraction**.



The Free-Body Diagram

Types of supports: Cables and Pulleys.

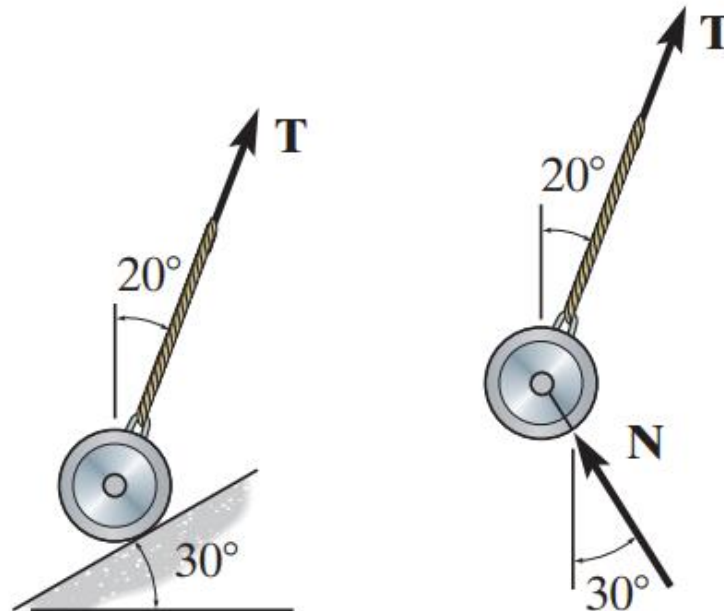
A cable can only support a tension or “pulling” force, and this force always acts in the direction of the cable.



The Free-Body Diagram

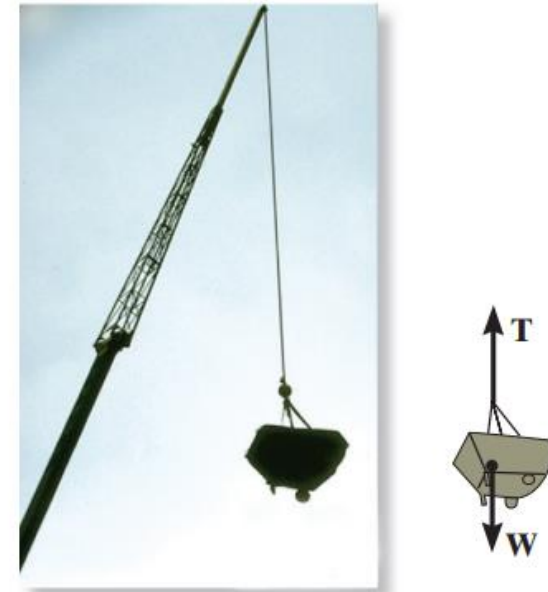
Types of supports: Smooth Contact.

If an object (or particle) rests on a smooth surface, then the surface will exert a force N on the object that is normal to the surface at the point of contact.



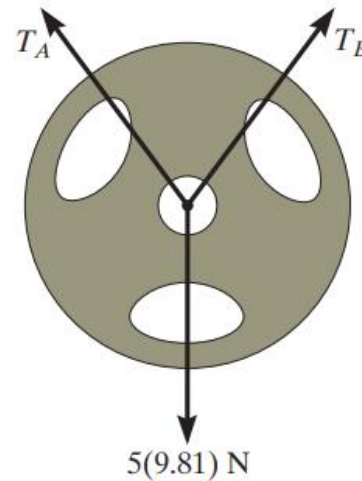
The Free-Body Diagram

1. Imagine the particle to be isolated or cut “free” from its surroundings. This requires removing all the supports and drawing the particle’s outlined shape.
2. Indicate on this sketch all the forces that act on the particle. These forces can be **active forces**, which tend to set the particle in motion, or they can be **reactive forces**, which are the result of the constraints or supports that tend to prevent motion.



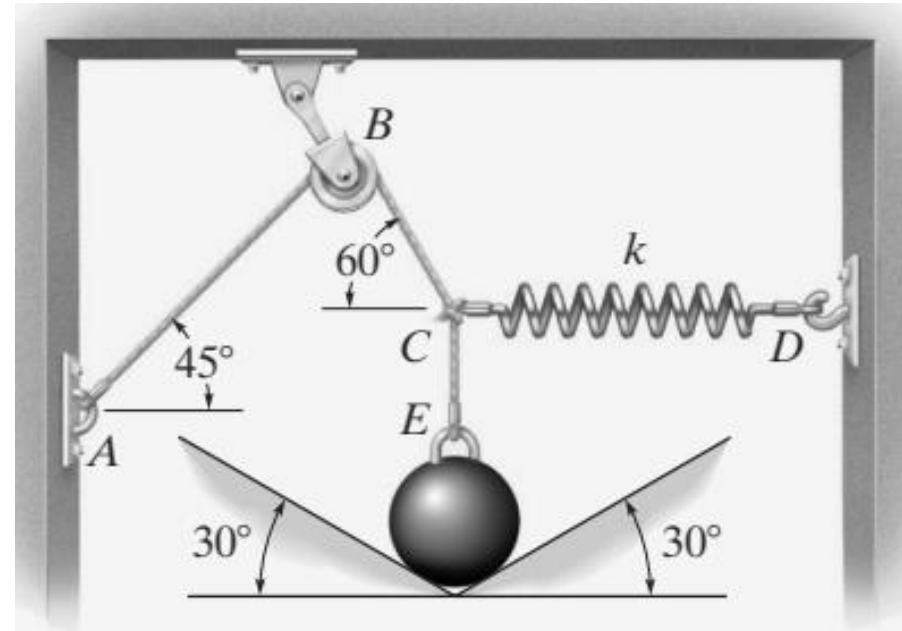
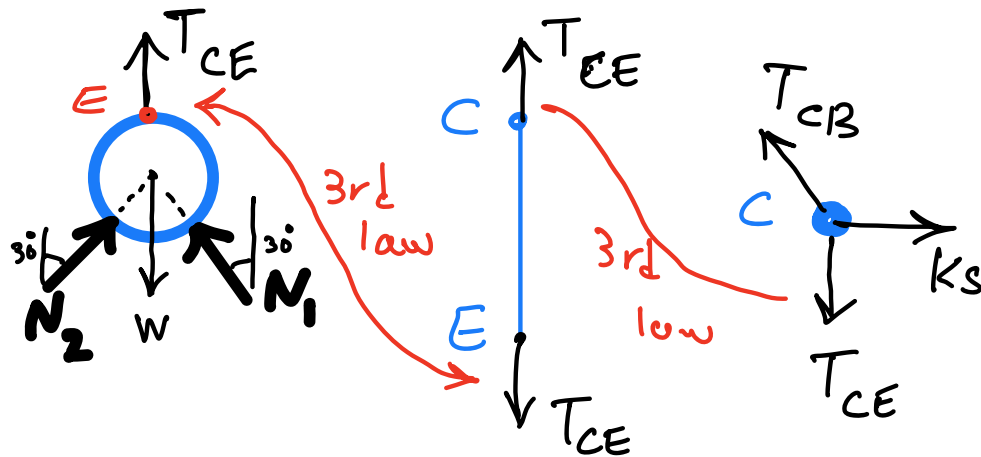
The bucket is suspended by the cable, and instinctively we know that for equilibrium, the force in the cable must equal the weight of the bucket. By drawing a free-body diagram of the bucket we can understand why this is so. This diagram shows that there are only two forces *acting on the bucket*, namely, its weight **W** and the force **T** of the cable, and since the resultant of these forces must be equal to zero, then $T = W$.

The Free-Body Diagram



The 5-kg plate is suspended by two straps *A* and *B*. To find the force in each strap we should consider the free-body diagram of the plate. As noted, the three forces acting on it are concurrent and therefore they can be applied to a fictitious particle at the center of the plate.

Example: The 6-kg sphere is supported as shown below. Draw a free-body diagram of the sphere, the cord CE, and the knot at C.

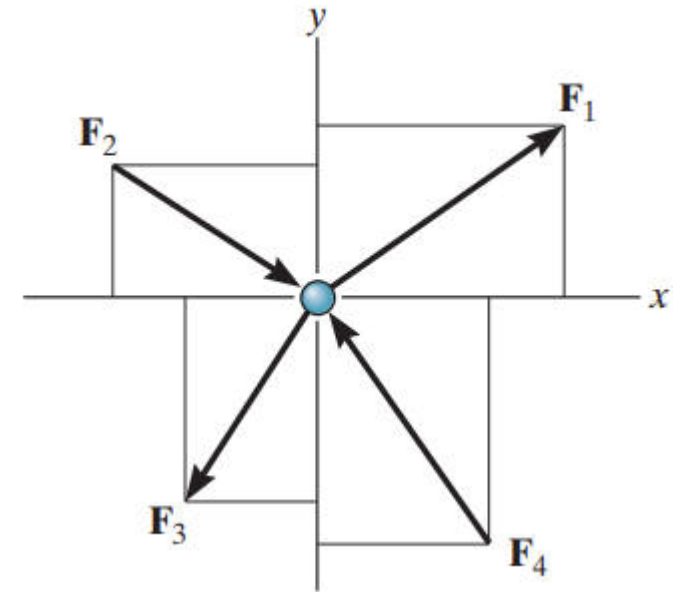


Coplanar Force Systems

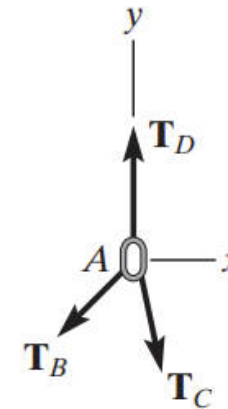
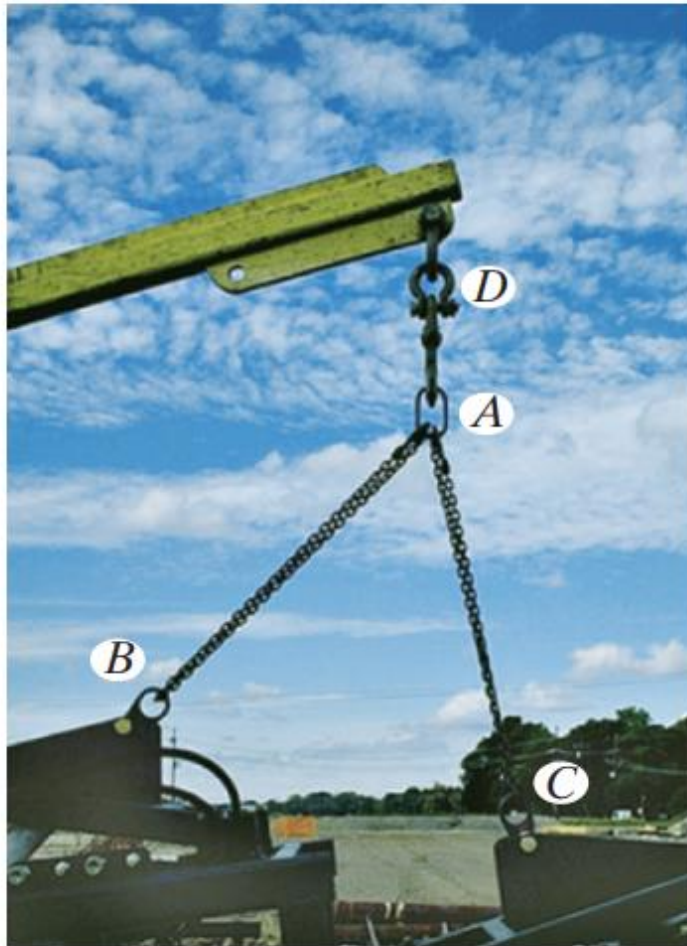
If a particle is subjected to a system of coplanar forces that lie in the x–y plane, then each force can be resolved into its **i** and **j** components. For equilibrium, these forces must sum to produce a zero-force resultant, i.e.,

$$\Sigma \mathbf{F} = \mathbf{0}$$

$$\Sigma F_x \mathbf{i} + \Sigma F_y \mathbf{j} = \mathbf{0}$$



Coplanar Force Systems

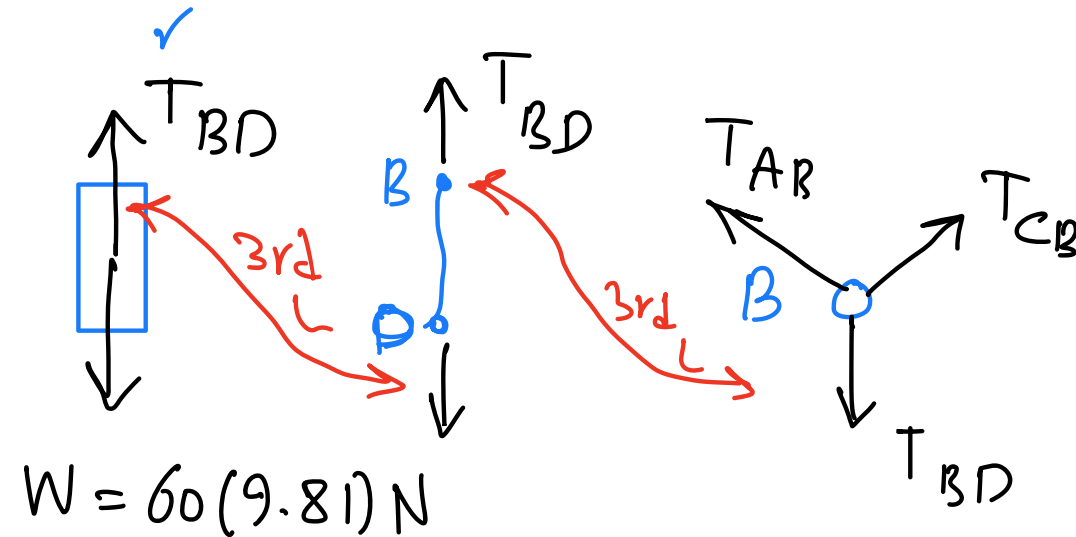


The chains exert three forces on the ring at A , as shown on its free-body diagram. The ring will not move, or will move with constant velocity, provided the summation of these forces along the x and along the y axis equals zero. If one of the three forces is known, the magnitudes of the other two forces can be obtained from the two equations of equilibrium.

Concluding Notes

- The first step in solving any equilibrium problem is to draw the particle's free-body diagram. This requires removing all the supports and isolating or freeing the particle from its surroundings and then showing all the forces that act on the particle.
- Equilibrium means the particle is at rest or moving at constant velocity. In two dimensions, the necessary and sufficient conditions for equilibrium require $\Sigma F_x = 0$ and $\Sigma F_y = 0$.

Example: Determine the tension in cables BA and BC necessary to support the 60-kg cylinder shown below.

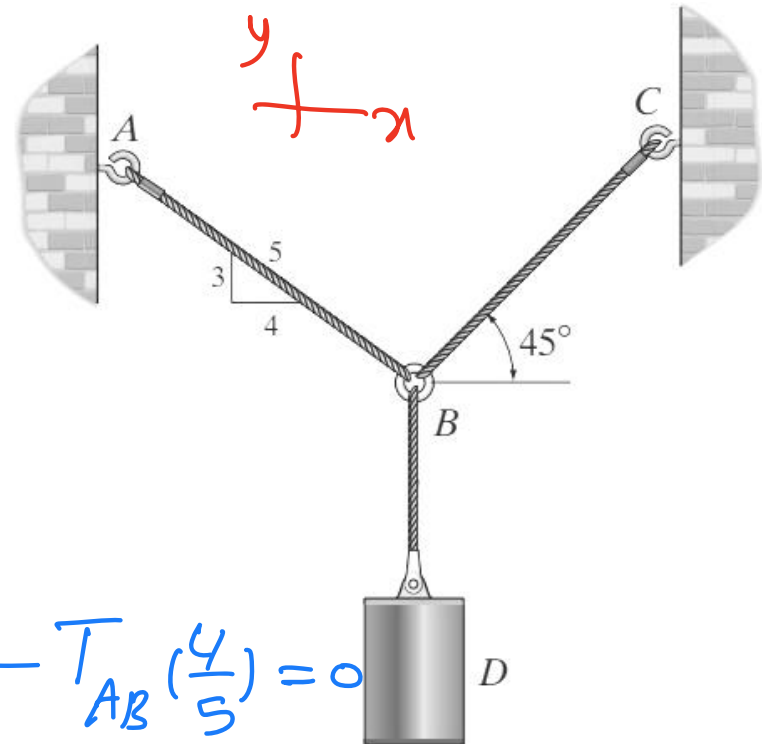


$$\sum F_y = 0$$

$$W = T_{BD}$$

$$\sum F_x = T_{CB} \cos 45^\circ - T_{AB} \left(\frac{4}{5}\right) = 0$$

$$\sum F_y = T_{CB} \sin 45^\circ + T_{AB} \left(\frac{3}{5}\right) - T_{BD} = 0$$

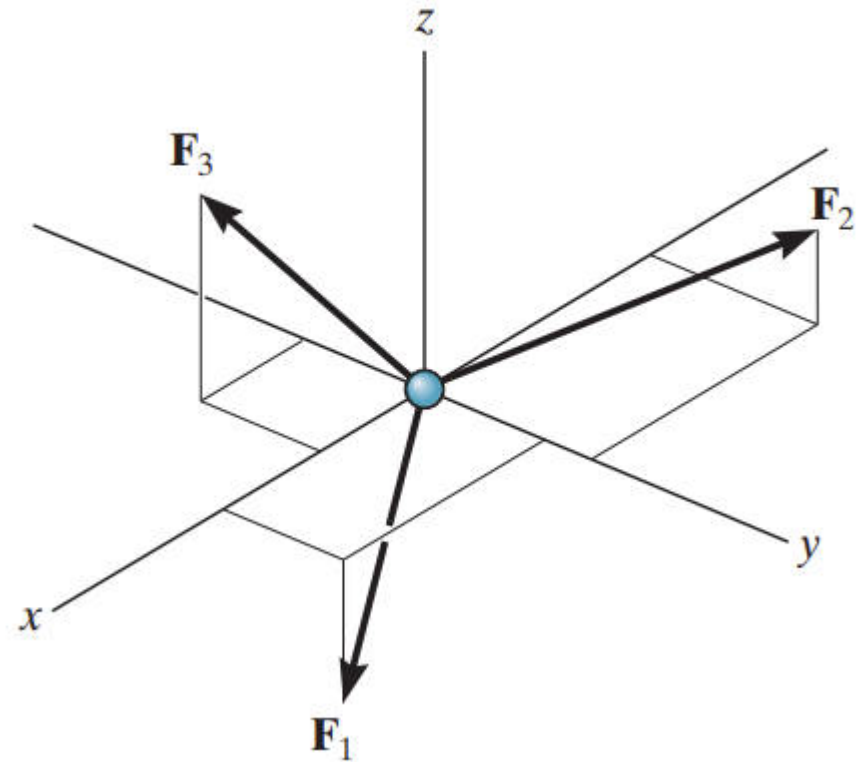


Three-Dimensional Force Systems

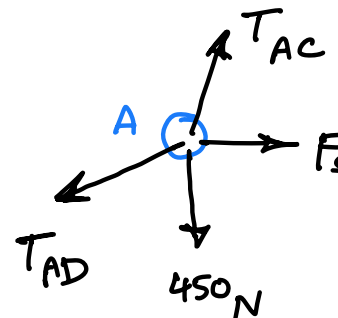
$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma F_z = 0$$



Example: If the 450-N load as shown below is supported by two cables and a spring having a stiffness $k = 8 \text{ kN/m}$, determine the force in the cables and the stretch of the spring for equilibrium. Cable AD lies in the x - y plane and cable AC lies in the x - z plane.



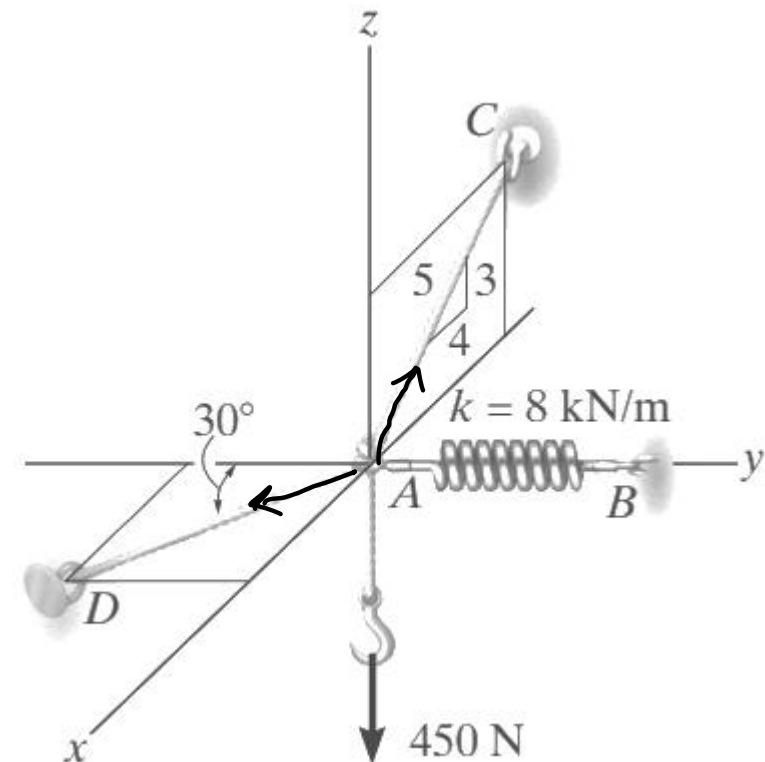
A free-body diagram of point A, represented by a blue circle. Four forces act on it: a tension force T_{AC} pointing up and to the right, a tension force T_{AD} pointing down and to the left, a horizontal force F_s pointing to the right, and a vertical force of 450 N pointing downwards.

$$\sum F_x = T_{AD} \sin 30^\circ - T_{AC} \left(\frac{4}{5}\right) = 0$$

$$\sum F_y = -T_{AD} \cos 30^\circ + F_s = 0$$

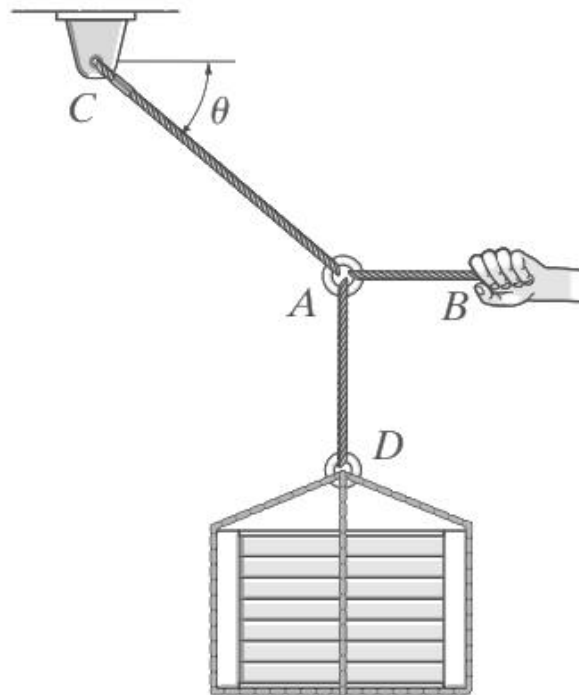
$$\sum F_z = T_{AC} \left(\frac{3}{5}\right) - 450 = 0$$

$F_s = 8 \text{ m} \rightarrow 8 \text{ m} \checkmark$



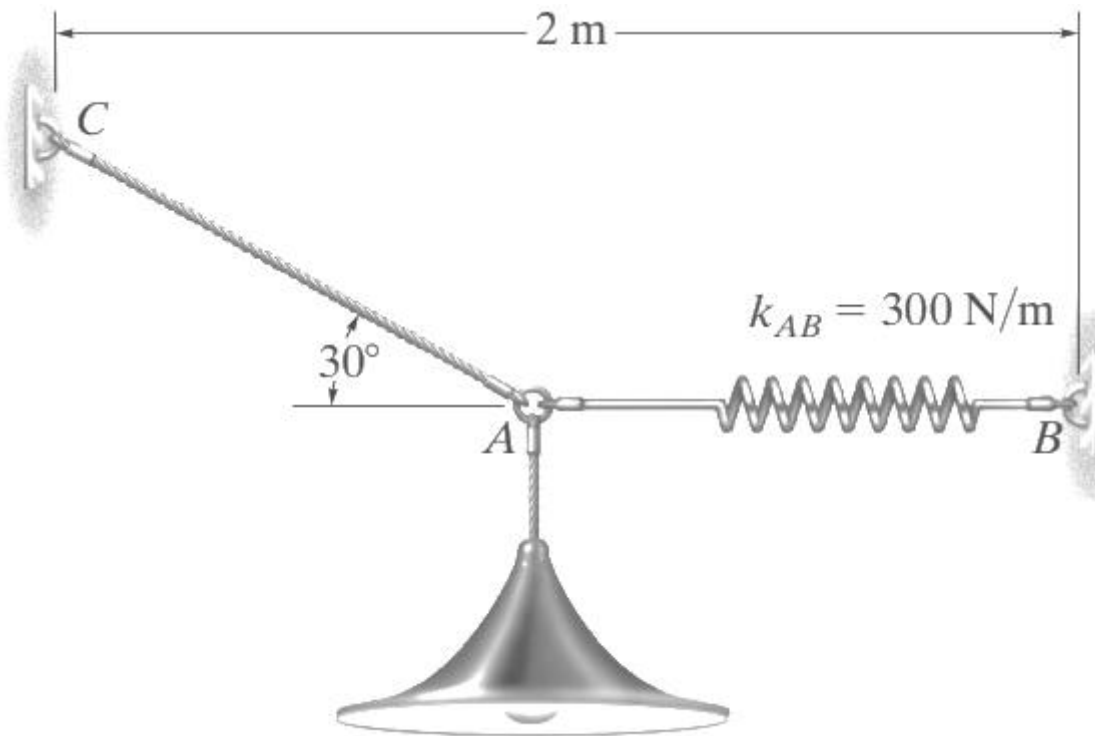
Practice at Home

1. Ropes AB and AC as shown can each support a maximum force of 10 kN. If AB always remains horizontal, determine the smallest angle θ to which the 200-kg crate can be suspended before one of the ropes breaks.



Practice at Home

2. Determine the required length of cord AC shown below so that the 8-kg lamp can be suspended in the position presented. The unstretched length of spring AB is $l_{AB} = 0.4$ m, and the spring has a stiffness of $k_{AB} = 300$ N/m.

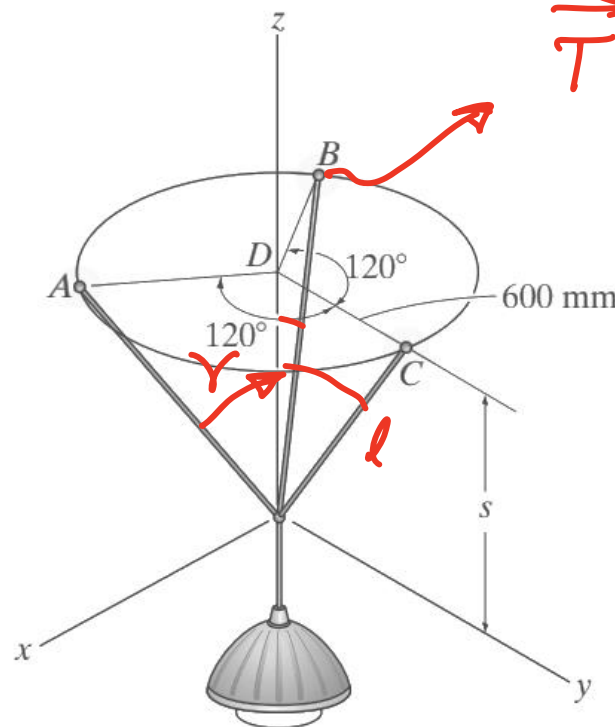


Practice at Home

3. The 10-kg lamp shown below is suspended from the three equal-length cords. Determine the smallest vertical distance s from the ceiling if the force developed in any cord is not allowed to exceed 50 N.

50 N
 $3T \cos \gamma$
 100 N

$\gamma = \gamma$



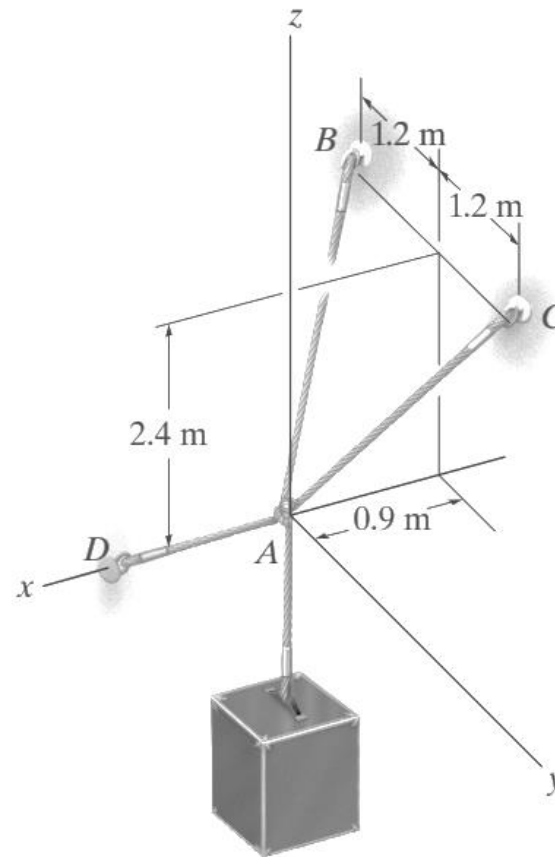
$\vec{T} = T(\cos \alpha \underline{i} + \cos \beta \underline{j} + \cos \gamma \underline{k})$

$l \sin \gamma = 600$

$l \cos \gamma = s$

Practice at Home

4. Determine the force in each cable used to support the 400-N crate shown below.



Practice at Home

5. Determine the tension in each cord used to support the 100-kg crate shown below.

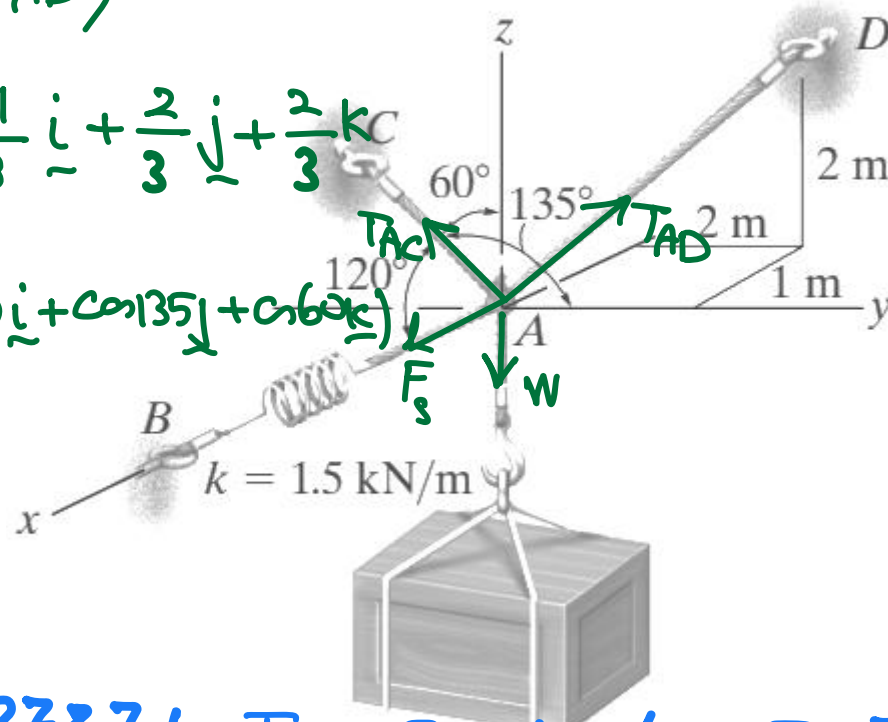
$$\vec{T}_{AD} = T_{AD} (\vec{u}_{AD})$$

$$\vec{u}_{AD} = \frac{\vec{r}_{DA}}{|\vec{r}_{DA}|} = \frac{-1}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{2}{3}\vec{k}$$

$$\vec{T}_{AC} = T_{AC} (\cos 120^\circ \vec{i} + \cos 135^\circ \vec{j} + \cos 60^\circ \vec{k})$$

$$\vec{F}_s = F_s \vec{i}$$

$$\vec{W} = -W \vec{k}$$



$$\sum F_x = 0$$

$$-\frac{T_{AD}}{3} + T_{AC} \cos 120^\circ + F_s = 0$$

$$\sum F_y = 0$$

$$\frac{2T_{AD}}{3} + T_{AC} \cos 135^\circ = 0$$

$$\sum F_z = 0$$

$$\frac{2}{3}T_{AD} + T_{AC} \cos 60^\circ - W = 0$$

$$T_{AD} = 878.7 \text{ N} \quad T_{AC} = 828.43 \text{ N} \quad F_s = 707.11 \text{ N}$$